

Solutions for Problem Set 5

Mathematics for Social Scientists

EXERCISE 1. Calculate the following definite and indefinite integrals: (*Some of these are from J. Gill, 2006, Ch5*):

(i) $\int_1^3 (x^2 + 4x + 3) dx$

(ii) $\int_{-1}^1 14 + x^2 dx$

(iii) $\int_2^4 e^y dy$

(iv) $\int (x^2 - x^{-\frac{1}{2}}) dx$

(v) $\int_{-1}^2 x\sqrt{1+x^2} dx$

(vi) $\int_1^2 \frac{1}{t^2} dt$

(vii) $\int e^{3xy} dx$

(viii) $\int 360t^{11} dx$

(i) $\frac{92}{3}$

(ii) $\frac{86}{3}$

(iii) $e^2(e-1)(e+1)$

(iv) $\frac{x^3}{3} - 2x^{\frac{1}{2}} + c$

(v) $\frac{1}{3}(5^{\frac{3}{2}} - 2^{\frac{3}{2}})$

(vi) $\frac{1}{2}$

(vii) $e^{3xy} \frac{1}{3y} + c$

(viii) $30t^{12} + c$

EXERCISE 2. Solve the indefinite integrals:

(i)

$$\int \frac{1}{\sqrt{x}} dx = \int x^{-1/2} dx = \frac{x^{(-1/2)+1}}{(-1/2)+1} + C = \frac{x^{1/2}}{1/2} + C = 2\sqrt{x} + C.$$

(ii)

$$\int e^{-4t} dt \stackrel{u=-4t}{=} \int e^u \frac{du}{-4} = -\frac{1}{4} \int e^u du = -\frac{1}{4} e^u + C = -\frac{1}{4} e^{-4t} + C.$$

(iii)

$$\int x\sqrt{x} dx = \int x \cdot x^{1/2} dx = \int x^{3/2} dx = \frac{x^{(3/2)+1}}{(3/2)+1} + C = \frac{x^{5/2}}{5/2} + C = \frac{2}{5} x^{5/2} + C.$$

(iv)

$$\int \frac{1}{x} dx = \ln|x| + C.$$

(v)

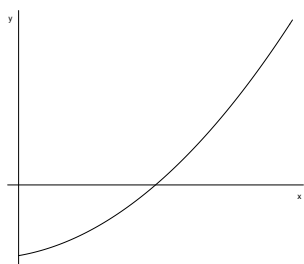
$$\begin{aligned} \int (2x^2 + x - 3) dx &= 2 \int x^2 dx + \int x dx - 3 \int 1 dx \\ &= 2 \cdot \frac{x^3}{3} + \frac{x^2}{2} - 3x + C = \frac{2}{3} x^3 + \frac{1}{2} x^2 - 3x + C. \end{aligned}$$

(vi)

$$\begin{aligned} \int \frac{(x^4 + 1)^2}{x^3} dx &= \int \frac{x^8 + 2x^4 + 1}{x^3} dx = \int (x^5 + 2x^1 + x^{-3}) dx \\ &= \frac{x^6}{6} + 2 \cdot \frac{x^2}{2} + \frac{x^{-2}}{-2} + C = \frac{1}{6} x^6 + x^2 - \frac{1}{2x^2} + C. \end{aligned}$$

EXERCISE 3.

Calculate $\int_0^2 (2x^2 + x - 3) dx$. Hint: Make a sketch of the function before.



At $x = 1$, the curve crosses the horizontal axis. In order to compute the total area "under the curve", compute $\int_0^1 (2x^2 + x - 3) dx + \int_1^2 (2x^2 + x - 3) dx$. Using the result from 2(v), we can write:

$$\begin{aligned}
 \int_0^1 (2x^2 + x - 3) dx + \int_1^2 (2x^2 + x - 3) dx &= \left| \frac{2}{3} + \frac{1}{2} - 3 - 0 \right| \\
 &\quad + \left| \frac{2}{3} \cdot 2^3 + \frac{1}{2} \cdot 2^2 - 3 \cdot 2 - \left(\frac{2}{3} + \frac{1}{2} - 3 \right) \right| \\
 &= \frac{11}{6} + \frac{4}{3} + \frac{11}{6} \\
 &= 5
 \end{aligned}$$

Alternatively, without absolute values the definite integral over $[0, 2]$ is

$$\int_0^2 (2x^2 + x - 3) dx = \left[\frac{2}{3}x^3 + \frac{1}{2}x^2 - 3x \right]_0^2 = \frac{16}{3} + 2 - 6 = \frac{4}{3}.$$

Remark: the definite integral counts area above the axis as positive and area below as negative. The total area adds the absolute values of the pieces (or splits at the zeros and takes absolute values).